

Outer Solar System Secular Kozai-Lidov Architecture & Perihelion Lifting

Batygin et al. (2020) — Secular Gravitational Coupling: Giant Planets + Planet Nine + Detached eTNOs

Kozai-Lidov Exchange

$$e(t) \leftrightarrow i(t)$$
$$q(t) = a(1 - e)$$

Detached Decoupling Boundary ($q = 40$ AU)

Neptune Scattering Corridor ($q \leq 36$ AU)

Neptune Orbit ($a_N = 30.1$ AU)

Perihelion Lifting

$q_0 = 33$ AU $\rightarrow q_{\text{max}} \approx 80$ AU

2012 VP113
 $q = 80.5$ AU, $a = 261$ AU

Sun

Giant Planets Field
 $\varpi_{\text{in}} > 0$

Planet Nine
($M \approx 5 M_{\oplus}$, $a \approx 460$ AU, $i \approx 16^\circ$)

Sedna (90377)
 $q = 76$ AU, $a = 506$ AU

Leleakuhonua
 $q = 65$ AU, $a = 1094$ AU

Apsidal Confinement: $\Delta\varpi = \varpi_{\text{TNO}} - \varpi_{P9} \approx 180^\circ$ (Anti-Aligned Configuration)